

PROMONTORY ROUTE RAILROAD TRESTLES
11 miles west of Corrine
Corrine Vicinity
Box Elder County
Utah

HAER NO. UT-64

HAER
UT-64

PHOTOGRAPHS

WRITTEN HISTORICAL AND DESCRIPTIVE DATA

Historic American Engineering Record
National Park Service
Rocky Mountain Regional Office
Department of the Interior
P. O. Box 25287
Denver, Colorado 80225

HISTORIC AMERICAN ENGINEERING RECORD PROMONTORY ROUTE RAILROAD TRESTLES

HAER No. UT-64

I. INTRODUCTION

Location: Spanning seven drainage channels in northern Great Salt Lake marshes, 11 miles west of Corriner, Utah

Quad: Public Shooting Grounds, Utah

Dates of Construction:	UT-64-A	1872-1936
	UT-64-B	1872-1940
	UT-64-C	1882-1940
	UT-64-D	1872-1940
	UT-64-E	1872-1940
	UT-64-F	1882-1940
	UT-64-G	1872-1940

UTM:	UT-64-A	12/388400/4606560
	UT-64-B	12/389860/4606000
	UT-64-C	12/390080/4605930
	UT-64-D	12/390700/4605690
	UT-64-E	12/391100/4605540
	UT-64-F	12/391980/4605205
	UT-64-G	12/393060/4604630

Present Owner: Southern Pacific Transportation are used as a Chevron Oil Company pipeline route and in part as a vehicular corridor. The trestles are to be demolished and replaced with earthen fill.

Significance:

The seven Promontory Branch railroad trestles were originally part of the first transcontinental railroad route constructed across the United States. As such they represent important remnants of this railroad which was instrumental in joining the nation together. They are also good examples of a class of small utilitarian wooden trestles constructed throughout the country during the latter half of the 19th century.

Historian:

Michael R. Polk, Sagebrush Archaeological Consultants, Ogden, Utah. October 1991.

Architect/Illustrator:

Allen C. Roberts, Cooper/Roberts Architects, Inc., Salt Lake City, Utah. September/November 1991.

Photographers:

Michael J. Bradshaw and Michael R. Polk, Sagebrush Archaeological Consultants, Ogden, Utah. March and April 1991.

II. TRANSCONTINENTAL RAILROAD HISTORY

A. NEED FOR RAILROAD

The construction of the Transcontinental Railroad is one of the most written about sagas in American history. A feat of such magnitude was unimaginable at the time, but it was a vital effort needed to physically bind the still-young nation together and permanently solidify its territory. The history of the Pacific Railroad (as it was originally called) development and construction began in the 1820's, about the time the first network of canals was completed in the eastern United States. During this period railroads began to spread throughout the East. By the 1850's they had spread into the Midwest to the Mississippi River. The desire to continue the expansion and to eventually span the continent with a Pacific Railroad grew and eventually became a great public debate.¹

This desire was not all grounded in a desire to connect the two parts of the country, rather promoters of such a railroad were foremost interested in and vocal about its commercial importance, even beyond this continent. As historian Robert Utley notes:

The settlement of the Oregon question in 1846, the discovery of gold in California in 1848, and the admission of California to statehood in 1850 swelled the population of the Pacific Coast. With commerce almost wholly dependent upon the long, slow journey around Cape Horn or across the Isthmus of Panama, both East and West foresaw a large and lucrative trade speeding by rail across the continent.²

The proponents saw the potential for diverting much of the Europe to Asia trade from ship to rail, but the most coveted objective was trade with China, Japan and other Asian areas.³

For the United States government these reasons were good, but most important in its final decision to actively promote and financially support this project were its potential effects upon domestic political and economic matters. From the government's perspective the railroad had the potential to more quickly end hostility with the American Indians and significantly reduce the expense and speed up transportation of mail and government supplies.⁴ In addition, the outbreak of the Civil War in 1861 made it clear that the bonds between California and the Union needed to be strengthened. Also, the Trent Affair, which almost caused another war between the United States and England, revealed the defenseless condition of the Pacific Coast.⁵ Though the construction of the Suez Canal in 1869 destroyed the potential value of the U.S. as a rail bridge for trade between Europe and Asia, the railroad was to fulfill all of its other expectations and more.

The real efforts to initiate construction began in the 1840's with promotion by a New York merchant in the China trade named Asa Whitney.⁶ His obsession with this project led him to write articles, lecture and talk with influential politicians. By the 1850's most nationally prominent politicians were in favor of such a plan with a measure of federal aid. But agreement could never be reached on an eastern terminus, a problem which was compounded by the lack of information about the merits of the different possible routes that could be used.⁷ This led to a series of fairly comprehensive Pacific Railroad surveys carried out by the Army Engineers between 1853 and 1855. The results of these surveys (two northern and two southern routes) were politically objectionable to both the Northerners and the Southerners. Thus, the issue remained unresolved. It took the Civil War, which removed the southern objections to a northern route, strong lobbying by Theodore Judah, a California railroad engi-

neer, and many eastern promoters to convince a beleaguered Congress to pass a bill in 1862 throwing the support of the United States government behind the effort.⁸ President Lincoln supported the effort and signed the bill into law on July 1, 1862. This act authorized the Central Pacific Railroad and Union Pacific Railroad and Telegraph Company to build the railroad from Omaha to Sacramento.

Construction of the railroad began in Sacramento on January 8, 1863 and in Omaha on December 2, 1863. The initial construction efforts were pathetic. The Civil War sent supply rates soaring and limited available labor in addition to drying up capital investment potential. By February 1864 only 18 miles of rail had been laid in California and none were laid out of Omaha until the spring of 1865. The adverse conditions of the time forced the railroads to ask for further assistance from the government, a request which was granted in the form of the Act of 1864. This act virtually doubled the resources available to the companies and insured the project's completion.⁹

B. CONSTRUCTION CHRONOLOGY

Nationwide Efforts

Between the years of 1864 and 1869 a total of 1,775 miles of rail were laid to complete the railroad link across the continent. This effort was perhaps the largest single construction project ever undertaken within the country. The task laid before both railroads was enormous, both logistically and financially. Following is a very brief chronology and description of the construction efforts, the obstacles faced and the solutions found to overcome them.

Despite the fact that the U.S. Government offered lucrative subsidies, these covered only half of the necessary capital needed to build the railroad. As a result, private investment was critical for both the Union Pacific and Central Pacific Railroads. Both companies devised the means to solve this dilemma by creating a number of indirectly held companies which carried out the construction work, but were not legally controlled by the Federal legislation which directed the efforts of the two main railroad companies.¹⁰

Construction of the transcontinental line was fraught with exceptionally difficult obstacles which sorely taxed the technological capabilities of the day. The Central Pacific spent four years surmounting the Sierra Nevada Mountains. The company faced the necessity of putting in deep fills and rock cuts, bridging deep canyons with trestles, and cutting through solid granite for 15 separate tunnels.¹¹ The Central Pacific did not reach Reno, Nevada until June 19, 1868. Reno lay 154 miles from the beginning of track in Sacramento. The 536 mile distance from there to Promontory Summit in northern Utah, however, was completed in less than 11 months.¹²

The Union Pacific did not have the same types of obstacles as the Central Pacific, but they too encountered great hardships. Track laying in the Platte River Valley of Nebraska was relatively easy, but the surveyors and construction workers soon encountered stiff opposition from Sioux and Cheyenne Indian war parties. Major skirmishes occurred in Nebraska and Wyoming between 1865 and 1867.¹³ The Union Pacific also encountered some difficult terrain in the Black Hills and, especially, in Echo Canyon and Weber Canyon in Utah.¹⁴

As both railroads approached Utah, it was well understood that negotiations needed to be completed with Brigham Young, President of the powerful Mormon Church, and former Governor of the Utah Territory, in order to complete the route. Young's desire was to see the railroad descend Weber Canyon and head south to Salt Lake City and around the south end of the Great Salt Lake.¹⁵ General Grenville M. Dodge, Chief Engineer for Union Pacific, however, preferred the northern route through the Promontory Mountains.¹⁶ The Central Pacific concurred in this assessment. Despite this basic disagreement, Brigham Young still agreed to provide needed labor and supplies to both the Union Pacific and Central Pacific Railroads. Not surprisingly, Young soon organized a new railroad company to build from Ogden to Salt Lake City once the transcontinental line was completed to Ogden. It became known as the Utah Central Railroad.¹⁷

Despite the many negotiations and bills passed by Congress in support of transcontinental railroad construction, neither the United States Government nor the Union Pacific or Central Pacific had ever developed a plan of where the lines should meet. As a result, for several months in 1869 surveyors and construction crews from both railroads continued their work on grading new lines well past any reasonable point of mutual connection. In early 1869 Central Pacific crews were grading as far east as Echo Summit near the Wyoming border while Union Pacific crews were working in the vicinity of Humboldt Wells, Nevada.¹⁸ The real purpose of the extra work carried out by each railroad was to gain both additional government subsidies and control of the potentially profitable Utah market.¹⁹ After much disagreement and negotiation, the two railroads agreed to join the line at Promontory Summit, Utah on May 10, 1869. This agreement was approved by Congress on April 10, 1869.²⁰

Construction of the Promontory Route

Central to the present investigation is the construction history of the rail route between Corinne and Promontory where the seven wooden trestles under study were built. The original construction of this part of the line was done by the Union Pacific Railroad between March 28 and May 8, 1869.²¹ The area in the mudflats west of Corinne and north of Bear River Bay posed few construction problems.²² This was one of the stretches of the route, however, where the Central Pacific built grade parallel to that of the Union Pacific during the period of intense competition between the two companies. In fact, because a meeting point for the two lines was never planned and there was a clear indication that the Union Pacific would be building west of Ogden before the Central Pacific could reach it, Leland Stanford of the Central Pacific chose to buy land and place surveyors and grading crews in this area in 1868.²³ As a result of this foresight, the Central Pacific was eventually able to leverage an agreement which placed their eastern terminus in Ogden.²⁴

Central Pacific began purchasing right-of-way and building their grade in the area in 1868. The Union Pacific did not even begin construction until February, 1869.²⁵ By March 1869 construction activity by both railroads was moving at a frenzied pace. A letter to the *Deseret Evening News* dated March 25, 1869 provides a firsthand account of this activity in the area between Corinne and Junction City (now known as Lampo Junction):

Work is being vigorously prosecuted...both lines running near each other and occasionally crossing. Both companies have their pile drivers at work where the lines cross the [Bear] river [near Corinne]. From Corinne west thirty miles, the grading camps present the appearance of a mighty army. As far as the eye can reach are to be seen almost a continuous line of tents, wagons and men.²⁶

The irritation that the close construction in this area caused the railroad owners was expressed by Leland Stanford of the Central Pacific in a March 14, 1869 letter to Central Pacific treasurer Mark Hopkins:

The U. P. have changed their line so as to cross us five times with unequal grades between Bear River and Promontory. They have done this purposely as there was no necessity for so doing. ...we shall serve notice for them not to interfere with our line and rest there for the present.²⁷

The principal contractors for the Central Pacific in this area were Ezra T. Benson of Logan; Lorin Farr, Mayor of Ogden; and Clancey W. West of Ogden. The Union Pacific contracted with the Mormon company of Sharp and Young.²⁸ All of these parties were hired through the recommendation of Brigham Young.

After the April meeting in Washington, D. C. at which the two companies agreed to join their railroads at Promontory Summit and the Union Pacific agreed to sell that portion of their line from Promontory to Ogden to the Central Pacific, the frenzied competition and redundant construction activities largely ceased. However, there was still much negotiation necessary before a price was agreed upon to transfer ownership of this portion of the transcontinental line. After two offers by Union Pacific were turned down by the Central Pacific, a sum of \$2,853,000 was agreed to on November 17, 1869 for a 48.5 mile section of line from Promontory to a point five miles west of Ogden.²⁹ Interestingly, the Union Pacific ended up selling the route for more than a million dollar loss based on its original construction cost. This price was agreed to, no doubt, due to the Union Pacific's desperate financial condition in 1869.³⁰

The Central Pacific Railroad continued to use most of the grade originally constructed by the Union Pacific between Promontory and Ogden, though they used their own grade on the east face of the Promontory Mountains (most prominently eliminating Union Pacific's "Big Trestle"). Until about December 1, 1869, however, the official date for transfer of the 48.5 mile section, Union Pacific still operated on their original line between Ogden and Promontory and transferred with the Central Pacific at Promontory Summit.

There was much criticism during the later stages of railroad construction activities about whether the work done by the two railroads was commensurate with requirements set forth in the Congressional Acts of 1862 and 1864. Specifically, there were complaints that the Union Pacific, in particular, was building a substandard railroad and receiving unjust compensation for it from the U.S. Government.³¹ In response to the mounting criticism President Grant appointed a special commissioner, Isaac N. Morris, to inspect the reportedly unacceptable parts of the Union Pacific. His report was quite unfavorable and called the railroad the worst over which he had ever traveled and actually dangerous in places. As part of his inspection tour, he described the condition of the line between Corinne and Lamo Junction (as part of the entire section between Ogden and the eastern base of the Promontory Mountains):

The Union Pacific road-bed['s]... width... is only the width of the tie, or 8 feet, sometimes a little over and sometimes a little under. ...the road-bed...is a mixture of dirt and sand...places are found where it is mostly dirt, and then portions are met which are chiefly, if not entirely of gravel.³²

Another commission of "Eminent Citizens" later inspected the route and found it acceptable as a first class railroad.³³ Nevertheless, Central Pacific apparently found it necessary to essentially rebuild the route after they acquired it in late 1869. This is partially supported by Morris' description of the railroad grade between Lampo Junction and Ogden.³⁴ Also, Southern Pacific Transportation Company bridge reports for seven trestles in the area show that five were reconstructed just after they acquired the line in 1872 and two others in 1882. These newer trestles were built to replace the original inadequate trestles built by the Union Pacific.³⁵

The transcontinental railroad continued to operate through Promontory Summit until the end of the 19th century. Just before the turn-of-the-century, E. H. Harriman gained control of both the Union Pacific and Central Pacific Railroads and began a massive upgrading of the route to make it a much more cost effective transportation system.³⁶ As part of a massive reconstruction effort on the Central Pacific portion of the road through Nevada, he ordered a lengthy trestle structure built across the northern end of the Great Salt lake to shorten the distance through Utah, straighten the track and, especially, to lessen the costs associated with operating over the Promontory Mountains.³⁷ The new line, called the "Lucin Cutoff", was completed on November 26, 1903, though freight traffic was not diverted from the old line until March 18, 1904 and passenger traffic until September 18, 1904.³⁸ It shortened the route by 44 miles and reduced monthly costs to the railroad by \$60,000.³⁹

The Promontory Route, or "Old Line" as it was often called, continued to act as a backup route in case of flooding along the lake route, to service wheat farmers along the Promontory Route and to continue to service the still operative tele-

graph line which paralleled that line.⁴⁰ In the early 1940's the need for iron and steel in the war effort prompted the government to approach the Southern Pacific Transportation Company, which had absorbed the Central Pacific Railroad in 1884, about donating the rails, spikes and tie plates to the war effort.⁴¹ They agreed, and in the first seven days of July 1942, the railroad tore up 123 miles of line between Corinne and Lucin.⁴² Interestingly, up until 1942 the railroad, with very little traffic, continued to regularly maintain the structures and grade on the Promontory Route. The 1941 Bridge Inspection Book shows that ties and trestle components were replaced on the line through the middle 1930's and even into 1941.⁴³

The Promontory Route existed for 73 years and was a vital component of the entire Union Pacific-Central Pacific transcontinental railroad for 35 years. The portion between Promontory and Ogden was a unique section of mainline because it was originally owned and built by the Union Pacific, but later acquired and upgraded by the Central Pacific. Because it was abandoned in the first half of the 20th century (1942) and lies in a very remote area of the country, the Promontory Route retains perhaps the best preserved examples of late 19th century railroad grades and railroad grade structures on the entire transcontinental route.

Promontory Route Trestles

Seven wooden trestles are included as part of the present study along the Promontory route between Corinne and Lampo Junction. These seven trestles are examples of several hundred structures which are known to exist along this route, many of which are currently protected within the Golden Spike National Historic Site.

These seven trestles under study (UT-64-A, B, C, D, E, F and G) were originally constructed as part of the Union Pacific's efforts to reach Promontory Summit during the construction of the transcontinental railroad. The stretch of line between Ogden and Promontory was likely built by the Mormon company of Sharp and Young during the period between March 28 and May 8, 1869.⁴⁴ John Sharp and Joseph A. Young, along with Joseph F. Nounan, were principals in a company which was recommended for the work by Brigham Young. They were particularly singled out for the Union Pacific's larger contracts on specialized work such as making bridge abutments and tunneling. As such, it is likely that this company built the original trestles on the line.

John Sharp, assistant superintendent of public works under Brigham Young, came to Utah in 1850 as a converted Mormon. He was first employed as a stone quarry worker and later became superintendent of the quarry. In 1854 he was ordained a Bishop and 10 years later attained the public works position.⁴⁵ After completion of the transcontinental railroad line, Sharp became superintendent of the Utah Central Railroad in 1871 and its president in 1873.⁴⁶

Joseph A. Young was the son of L.D.S. President Brigham Young. He was originally appointed to oversee all surveying work on the Union Pacific construction over the Wasatch Mountains in 1864.⁴⁷ In the same year that Young

was completing the contracts for the Union Pacific (1869) he became general superintendent of the newly formed Utah Central Railroad.⁴⁸ Joseph F. Nounan was a non-Mormon businessman from Salt Lake City.⁴⁹

While a detailed description of the original seven trestles probably does not exist, during Morris' 1869 inspection tour of the line, he gave some brief information about them:

There are about sixteen culverts in the valley...These culverts, with two exceptions, which had sandstone abutments, are constructed of pine timber. Substantial stone abutments should be substituted as soon as possible. They are evidently required by public interest and the standard of construction.⁵⁰

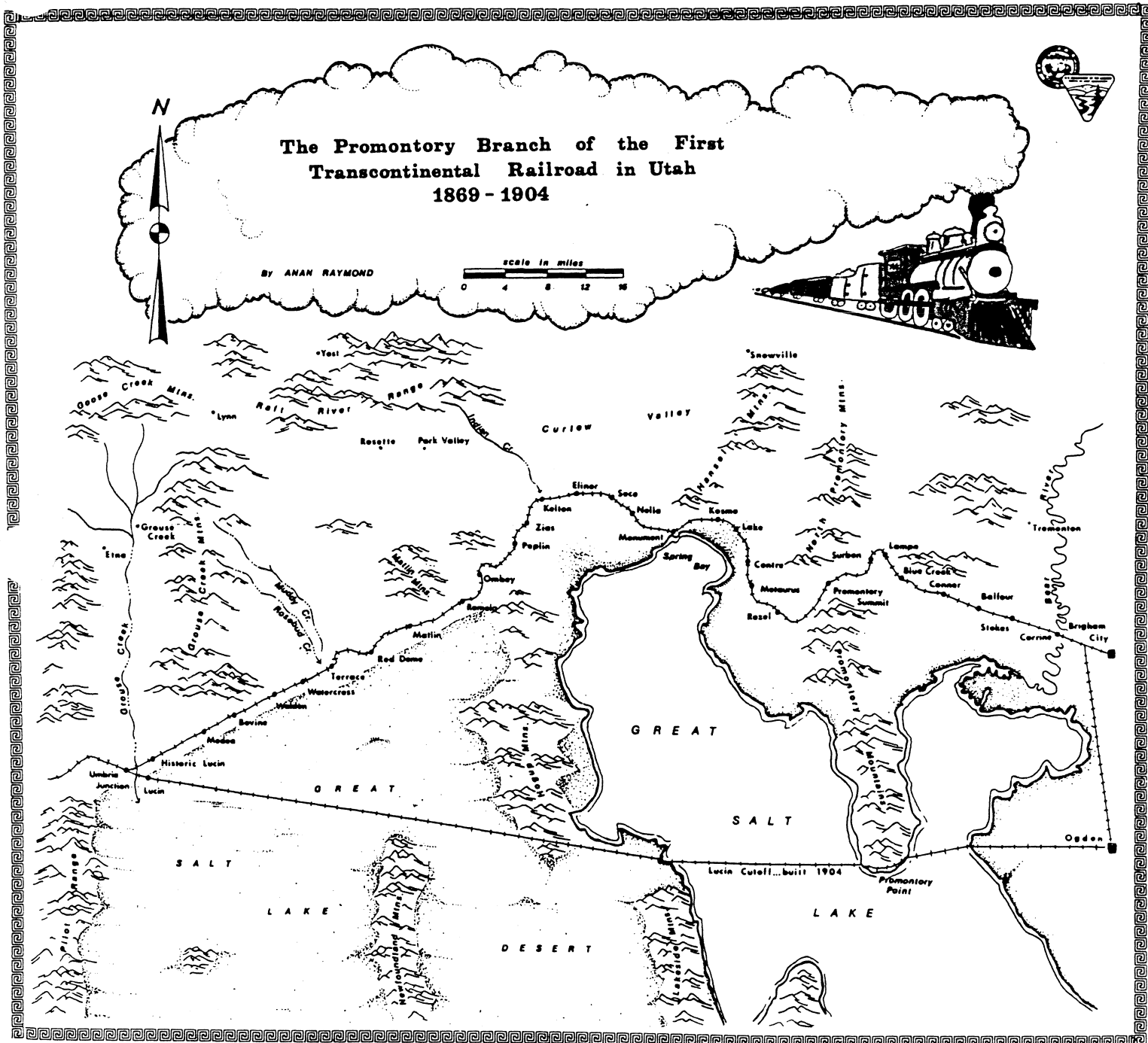
In 1876 the U.S. Government directed members of the U.S. Corps of Engineers to survey the entire completed transcontinental for payment purposes. This survey involved exact measurements of the entire line and identification of all structures. In his report on the section between Ogden and Battle Mountain, Nevada, Captain James F. Gregory referred to six of the trestles under study here as "culverts" and one as a trestle.⁵¹ Culvert is a term which may have been broadly used in referring to single span stringers or short length trestles. Following is a list of those "culverts" and one trestle which likely correspond to those still extant today:⁵²

HAER Designation	1876 Survey Designation	Distance from Ogden, Utah
UT-64-G	Culvert 285	34 miles, 172 feet
UT-64-F	Open Culvert	34 miles, 4159 feet
UT-64-E	Culvert 282, open	35 miles, 1151 feet
UT-64-D	Culvert 281, open	35 miles, 3300 feet
UT-64-C	Open Culvert	35 miles, 5274 feet
UT-64-B	Culvert 279, open	36 miles, 640 feet
UT-64-A	Trestle 277, (35 feet)	37 miles, 472 feet

The replacement trestle structures were, presumably, all built by Central Pacific Railroad construction crews. This information is surmised from the 1920 and 1941 maintenance records for the line. Original construction records are either not available or no longer exist.

All seven trestles under study are low, simple wooden utilitarian structures which were placed along the right-of-way to provide openings for water to pass under the railroad grade. They include four simple single span stringers (UT-64-B, C, F and G), two two-span trestles (UT-64-A and E) and one five-span structure (UT-64-D). All of the structures are made of large, milled and treated timbers. Two of the trestles (UT-64-F and G) contain remnants of the unshaped, untreated juniper tree pilings from the original trestle structure built by the Union Pacific in 1869. They were still visible adjacent to the present trestle pilings.

All of the structures have been extensively remodeled through time. Information available from the 1920 and 1941 Bridge Inspection Books for the Salt Lake Division of the Southern Pacific show changes that were made during the early 20th century.⁵³ It is likely that other changes occurred, but the books showing earlier work are not currently available for inspection. These structures served a very heavily trafficked railroad between 1869 and 1904 before the Lucin Cut-off was completed, so it is likely that many other changes not shown in the two maintenance records were carried out during that 35-year time frame. Regardless, these structures remain good examples of simple utilitarian grade structures of this period and likely saw only minimal changes after heavy rail traffic ceased over the line in 1904.



Taken from: Raymond, Anan S. and Richard E. Fike. *Rails East to Promontory, The Utah Stations*. Cultural Resource Series No. 8, Bureau of Land Management, Utah, 1981.

Southern Pacific Transportation Company

Southern Pacific Building • One Market Plaza • San Francisco, California 94105

C. J. BURROUGHS
CHIEF ENGINEER

IN REPLY PLEASE REFER TO

March 25, 1991

110-3

Mr. Michael R. Polk
c/o Sagebrush Archaeological Consultants
4263 Monroe Boulevard
Ogden, Utah 84403

Dear Mr. Polk:

This is in reply to information you requested in your letter of February 10, 1991, on seven trestles located on Southern Pacific's former Promontory Branch. The structures are (were) known as follows:

Your ID	SPT ID	Length	Type	Engr. Station	Date Built
A	788B	28 ft.	Open Deck Pile Trestle	2542+71	1872
B	789B	13"	"	2491+94	1872
C	789C	10"	"	2485+41	1882
D	790A	80"	"	2466+30	1882
E	790B	32"	"	2444+49	1872
F	790C	10"	"	2421+49	1882
G	791B	16"	"	2381+69	1872

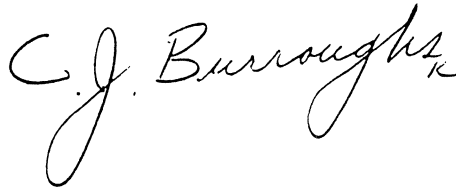
I have enclosed a page from the 1920 bridge inspection book which shows all seven trestles plus three culverts. The "year built" is the year the structure was built by (then) Central Pacific forces. The original structures were all built by Union Pacific forces in 1869 and we have no information as to their description. There is a bridge inspection book for each year between 1920 and 1942 when the Promontory Branch was abandoned. They are in a warehouse and I have requested the 1941 book for the former Salt Lake Division be sent to me. If there is any information that would be of help to you I will send you a copy of the report.

In looking at the photographs of each trestle that you sent with your letter it is obvious that the wood materials shown for all trestles is more recent than 1872 or 1882. You will note on the 1920 bridge inspection page that Structure 789B had four untreated piles and a new deck installed in 1920. It is certain that the other six trestles were renewed at various times but it would be too time consuming to go through the inspection books for each year from about 1907 to 1942.

- 2 -

I hope this information will be helpful to you.

Very truly yours,

A handwritten signature in cursive script, reading "C. J. Burroughs". The signature is written in dark ink and is positioned to the right of the typed name "C. J. Burroughs".

Enc.

LDF:sw

1920 Bridge Report					253
	NO.	LENGTH	KIND OF STRUCTURE	HT TO B. OF R.	YEAR BUILT
A	788 A		18x18 REDWOOD BOX 25'	6	1910
	788 B	28	OPEN DECK (4-7x16 Strs) 4-Untreated piles Hdr1	6	1872
	789 A		2x3 REDWOOD BOX 10'-0"	3	1879
B	789 B	13	OPEN DECK (3-7x16 Strs) 4unt-piles New dk. 1920	6	1872
C	789 C	10	OPEN DECK (3-7x16 Strs) 4-Untreated piles	6	1882
D	790 A	80	OPEN DECK (4-7 xl6 Strs) 4-Untreated piles Ties bad str & poor Repair	6	1872
E	790 B	32	OPEN DECK (4-7x16 Strs) 4-Untreated piles (it will) 90 years	8	1872
F	790 C	10	OPEN DECK (3-7x16 Strs) 4-Untreated piles	7	1882
	791 A		Stone box	8	1869
G	791 B	16	Covered up OPEN DECK (4-7x16 Strs) 4-Untreated piles str & ties bad renew-	8	1872

Page three of a letter from C. J. Burroughs, Chief, Engineering Department of Southern Pacific Transportation Company to the author on March 25, 1991. Southern Pacific Transportation Company, Salt Lake Division, Bridge Inspection Book, 1941. On file at the Southern Pacific Transportation Company, San Francisco, California.

Southern Pacific Transportation Company

Southern Pacific Building • One Market Plaza • San Francisco, California 94105

C. J. BURROUGHS
CHIEF ENGINEER

IN REPLY PLEASE REFER TO

April 3, 1991

E 110-3

Mr. Michael R. Polk
c/o Sagebrush Archaeological Consultants
4263 Monroe Boulevard
Ogden, Utah 84403

Dear Mr. Polk:

Further my letter to you of March 25, 1991 concerning data on seven old trestles on our former Promontory Branch.

I have reviewed the 1941 bridge inspection book and enclose copies of two pages relating to the trestles in which you are interested. You will note that trestle 788B, your A, which by 1941 is referred to as (mile) 788.83, had ties replaced (removed) in 1930 and 2 stringers renewed in 1933 and 2 more stringers in 1936. The other trestles have all been renewed at least in part. An explanation of abbreviations used follows:

Dk	-	Deck (the portion above the caps on the pile bents)
Strs	-	Stringers (the longitudinal members resting on the caps and on which the ties are placed).
Untr	-	Untreated wooden material
Rwd	-	Redwood timber
C.S.33-	-	A common standard plan
S.H.	-	Second hand
Blk	-	Bulkhead

Note that trestle 789C is shown in 1920 as having 7" x 16" stringers while the 1941 book shows 8" x 18" stringers. The renewal was done sometime between 1920 and 1941 but a review of many books would be needed to determine the exact year.

Again, I hope this information will be useful to you.

Very truly yours,



Encls.

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8-21-41

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NO.	LENGTH	KIND OF STRUCTURE	HT. TO B. OF R.	YEAR BUILT
787E		18"x18" Stone Box	6'	1869
787.96		Good		
788A		18"x18" Rwd. Box 2 Tr.	6'	1910
788.24	ONE	TRACK OFF 1941	Good	
		CONNOR		
⑦ 788B	28'	Open Dk. (4-7"x16" Strs.)	6'	1872
788.83		4-Pile Bents Untr.	TIE PLATES	
		TIES 1930	STRS 3-5 1933	
		PHOTO 1166	" 4-8 1936	
		Good		
789A		2'x3' Rwd. Box	3'	1921
789.18		Good		
⑦ 789B	13'	Open Dk. (3-7"x16" Strs.)	6'	1872
789.80		4-Pile Bents Rwd.	TIE PLATES	
		TIES 1940	Good	
⑦ 789C	10'	Open Dk. (3-8"x18" Strs.)	6'	1882
789.92		4-Pile Bents Untr.	CAP 1930	1938
		STRS 1938	TIE PLATED 1938	TIES 1941
		Good		
⑦ 790A	80'	Open Dk. (C.S. 33 Strs.)	6'	1872
790.28		4-Pile Bents Untr.	TIE PLATED 1938	
		DK. 1921.	REPAIRED 1930	
	BENT 4	PILE 4 SPURED:	CAP # 1 1940	
	" 5	CAP 1936		
		Good		
⑦ 790B	32'	Open Dk. (4-7"x16" Strs.)	8'	1872
790.69		4-Pile Bents Untr.	CAP 1933	
		3 TIES 1940	" 3 1936	
		Good		

Page two of a letter from C. J. Burroughs, Chief, Engineering Department of Southern Pacific Transportation Company to the author dated April 3, 1991. Southern Pacific Transportation Company, Salt Lake Division, Bridge Inspection Book, 1920. On file at the Southern Pacific Transportation Company, San Francisco, California.

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375-

No.	LENGTH	KIND OF STRUCTURE	HT TO B OF R	YEAR BUILT
790 C	10'	Open Dk. (3-8"x17" Strs.)	7'	1882
791.13		4-Pile Bents Untr.	STRS S.H. 1933	
TIE PLATED	3	TIES 1940	GOOD	CAPS 1934
791 A		2'x2' Stone Box	7'	1869
791.31			GOOD	
791 B	16'	Open Dk. (C.S. 33 STRS 1933)	8'	1872
791.88		4-Pile Bents Untr.	TIE PLATED	
4th STR 1938		4 STRUTS BLK TO BLK 1938	2 TIES 1940	
2 PILES		IN EACH UPSTREAM WING OF BLKS		
			GOOD	
791		24"x30" Stone Box	8'	1869
792.13			GOOD	
792 A	9'	Open Dk. (3-7"x16" Strs.)	9'	1884
792.69		4-Pile Bents Untr.	TIE PLATED	
4 STRUTS BLK TO BLK 1938			CAP 2 1938	
		TIES 1940 D.F.P.	GOOD	
792 B		2'x2' Stone Box	8'	1869
793.04			GOOD	
793.4		18"x36" Stone Box	10'	1869
793.49			GOOD	
		— BALFOUR —		
793 B		2'x2' Stone Box	8'	1869
794.27			GOOD	
794 A		2'x2' Stone Box	7'	1869
794.53			GOOD	
794 B		2'x2' Stone Box	6'	1869
794.67			GOOD	
794 C		24"x30" Stone Box	9'	1869
794.77			GOOD	
794 D		24"x30" Stone Box	11'	1869
795.08			GOOD	
795 A	14'	Open Dk. (3-7"x16" Strs.)	14'	1884
795.40		4-Pile Bents Untr.	TIE PLATED	
CAPS ON OLD BENTS 1933		6 TIES 1940		
THERE ARE TWO BENTS AT EACH END - THE OLD ORIGINAL				
BENTS AND A BENT IN FRONT OF THEM.				
THE BLK WINGS WERE REINFORCED 1938 WITH 2 X 3				
UNTR PILES EACH WING - SEVERAL STRUTS CRUSHING				
PUT IN MORE STRUTS				
GOOD FOR YEAR OR TWO				

Page three of a letter from C. J. Burroughs, Chief, Engineering Department of Southern Pacific Transportation Company to the author dated April 3, 1991. Southern Pacific Transportation Company, Salt Lake Division, Bridge Inspection Book, 1920. On file at the Southern Pacific Transportation Company, San Francisco, California.

III. ENDNOTES

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52. Ibid; Trestles UT-64-A, B, D, E, and G had all been replaced by the time this survey was conducted (1876). However, trestles UT-64-C and F were, presumably, still the same structures originally built by the Union Pacific in 1869.
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